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WINTER INTERNSHIP

Microbiology techniques and
Aquatic Toxicology

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PREFACE

This report summarizes the work and learning undertaken during a one-month winter internship under the supervision of **Dr. Reshma Sinha**. The focus of the internship was on standard techniques in **microbiology** and introduction to **aquatic toxicology**.

Over the course of the month, I was introduced to routine microbiological procedures such as sterilization, DNA isolation, gel electrophoresis, staining, and spectrophotometry. On the aquatic toxicology side, the work also involved understanding how pollutants can affect aquatic organisms.

Given the short duration, the internship was intended to provide an overview of experimental practical setup. The laboratory protocols, safety practices, and environmental relevance of microbial and toxicological studies are up to the standards.

I would like to thank **Dr. Reshma Sinha** for allowing me to participate in this internship and for her guidance throughout the process.

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5. ADDITIONALS: MICROTOME, PHTOCHEM. EXTRACTION & SOFTWARE PROGRAMS

1. A BRIEF OVERVIEW OF OXIDATIVE STRESS AND MICROPLASTICS IN AQUATIC TOXICOLOGY

1.1 Overview of Oxidative Stress

1.1.1 Oxidative Stress

1. Reactive Oxygen Species, ROS Generation
2. Imbalance between ROS and Antioxidants
3. Macromolecular Damage
4. ROS Cellular Dysfunction and Signalling Disruption
5. Tissue damage and disease progression

1.1.2 Steps involved:

1. Superoxide formation
2. SOD, Superoxide Dismutase activity to Hydrogen Peroxide
3. CAT, Catalase and GPx, Glutathione peroxidase detoxification
4. Lipid Peroxidation

1.2 AQUATIC TOXICOLOGICAL VIEW:

1.2.1. Why choose *Danio rerio* (zebrafish)? Model Organism

- Easy to maintain
- Size: 1.7-3.8 cm
- Three to five years of life; 84% of the genome is like humans.
- Infinity movements in stress conditions.

1.2.2. Oxidative stress: Consequences and Responses

- *Danio rerio* is given less than LC₅₀ concentrations of PA+++ in water to determine long-term impacts on the fish.
- Damage to Liver, Kidney, Brain and blood.
- Identification of LPA by MDA, Malondialdehyde (marker of LPA) - TBARS, Thiobarbituric Assay forms pink chromogen measured spectrophotometrically
- Protein oxidation Carbonyl content (Spectrophotometric assay: Beer Lambert's law) 8OHdG by ELISA

1.2.3. Blood smear to check oxidative stress

- Prepare blood smear to check oxidative stress.
- Pour blood, stick using other slide and put it in dye Wright Giemsa.
- Leave it for 15-20 minutes and then observe it under the microscope to check incomplete cells. To consider a damaged cell, 70 percent of its part must be visible.

1.3 Microplastics indicator:

Ophichthys (snaketail dragonfly), A macroinvertebrate. Their nymphs inhabit along the shores of clear flowing water streams, well oxygenated water having substrates of fine gravels. All organic material is degraded by 30 percent H₂O within 72 hrs and micro plastics residues are left.

2. DNA ISOLATION

2.1 Autoclave

Autoclave is a device which operates at high temperature and pressure to kill microorganisms and spores effectively to maintain the sterile conditions. Make sure that water levels are up to the mark before operating it. 121 degrees Celsius, 15 psi and steam sterilize for at least 30 minutes.

To maintain sterile conditions:

1. Glassware is placed at high temperature.
2. Plasticware in autoclave device.
3. Buffers in sterile and non-reactive environment.

2.2 Buffer Preparation

2.2.1. Preparation of Lysis Buffer

1000µL of .5M EDTA	Intercalating agent
1000 µL of 10 % SDS	Anionic detergent
500 µL of 1M Tris Cl	Maintains pH
500 µL of 250 Mm NaCl	Osmolarity

i. TE Buffer

40L of .5 M EDTA

200 of 1 M Tris HCl

20 ml H₂O

At last autoclave, filter sterilize and store the buffer at room temperature.

2.2.3. Other chemicals required:

3M CH₃COONA: Neutralises charge, facilitate precipitation and remove contaminants.

Proteinase K: 25 mg in 1 ml H₂O

CHCl₃: isoamyl alcohol in (24:1)

Ethanol

2.3 Protocol:

1. Sample collection: 20, 50-150 µL of LB
2. Homogenise , then water bath at 37 degrees for 30 min.

3. Add proteinase K 5 μ L
4. Water bath 55-65 °C for 60 min
5. 100 μ L of CH₃ COONa and ice bath for 60 min
6. Centrifuge
7. Supernatant transfer
8. Equal amt of phenol, vortex, equal amt of CHCl₃: isoamyl alcohol, vortex
9. Centrifuge at 8000 rpm for 10 min at 4 degrees
10. Aqueous layer, prechilled ethanol, invert
11. Incubate for night
12. Centrifuge, drain supernatant
13. 200 μ L of ethanol room temp
14. Centrifuge
15. Remove supernatant
16. Put in TE buffer

3. Separating DNA Fragments using Gel Electrophoresis:

Gel electrophoresis is used to separate DNA fragments based on size. An electric current onto the DNA samples through a gel after they have been loaded into wells at one end. Due to their negative charge, DNA fragments move towards the positive electrode. Smaller molecules will move more quickly than larger ones as negatively charged molecules move away from the electric current's negative pole.

3.1. Gel Electrophoresis: 50-150 V for 45-90 minutes

1. Acetic Acid EDTA Working solution
2. .8g agarose for 50 mg TAE buffer
3. 1mg EtBr: Intercalating agent; helps in visualising DNA
4. Loading Buffer

3.2. Loading gel (Standard protocol):

6X DNA Loading Buffer (for gel electrophoresis)

- Ficoll 400 – 2.5%
- EDTA – 11 mM
- Tris-HCl – 3.3 mM (pH 8.0 @ 25 °C)
- SDS – 0.017%
- Bromophenol blue – 0.015%
- Xylene cyanol – 0.015%
- Bring up to 50 mL with dH₂O
- Aliquot and store at room temperature (RT).

1X TAE Buffer (Working Solution)

- To prepare **10 L** of 1X TAE from 25X stock.
- Mix **400 mL** of 25X TAE stock with **9.6 L** dH₂O

50X TAE Stock (for 1 L)

- Tris base – 242 g
- Glacial acetic acid – 52.1 mL
- Na₂EDTA·2H₂O – 37.2 g
- Add dH₂O to 1 L total volume

Load the sample with equal amount of loading gel and run it for 60 minutes. Sharp separate DNA bands will be visible.

4. SPECTROPHOTOMETER

Principle: **Transmittance, Reflection, Absorbance**

BEER LAMBERT'S LAW: The absorbance is directly proportional to the concentration and path length of the sample.

$A \propto cl$

$A = \epsilon cl$; where ϵ is molar absorptivity.

Cuvettes are used to put the sample.

Double beam or Single beam

The absorbance of DNA occurs at 260 nm.

Sample purity is determined by two ways i.e. 260/230 and 280/260 absorbance ratios:

- **260/280 ratio:** A good estimate of sample purity. For RNA, it should be around 2. Lower values may indicate contamination.
- **260/230 ratio:** Measures contaminants that absorb at 230 nm, such as proteins or phenol.

Double Beam Spectrophotometry:

- **Reference beam:** The standard so that the other things can be neglected like buffer.
- **Sample beam:** The reference beam traverses the sample medium to reflect the absorption of said sample.

Used to determine the integrity (quality) and quantity of DNA.

5. ADDITIONALS: MICROTOME, PHTOCHEM. EXTRACTION & SOFTWARE PROGRAMS

5.1 How to prepare Tissue blocks of Zebra Fish and trim using Microtome?

Fixative: 4% Paraformaldehyde (PFA) in PBS or 10% Neutral Buffered Formalin (NBF) (Not sure what exactly was used but have observed the use of the same buffers in the fixation of mouse tissues). Leave for 24-48 hrs for fixation and wash to remove residues with ethanol.

Put the tissues in molten paraffin (58–60 °C) for 2–3 hours in a block. Label the block and use microtome to start trimming the layers and set the thickness as desired.

5.2 Phytochemical Extraction:

Grind the powder for extracting and wrap it in filter paper put it in the Soxhlet, connect it to a volumetric flask keep the water flow slowly, and start the burner at 50-60 degrees Celsius.

The solvent starts to lighten up in a day.

Then put it in a rota evaporator

Condenser at 3

Temperature 50.1 degrees

Dec pressure by 5 units

5.3 POPs Extraction from soil samples of different locations using ZnCl_2 and **NaCl**.

5.4 Use of Software:

Zotero, PRISMA, Excel for Data collection and Database explorations.

